

Physical Chemistry of Materials

Lecture 17

2차 중간고사 안내 및 복습



2차 중간고사 안내

- 시간: 11월 19일(화) 오전 10:30~11:30
- 장소: 화학관 414호(홀수 학번), 112호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

고체(10강)

결정 진동: 고전적인 그림(11강)

포논(12강)

고체의 전자 구조(13강)

자성(14강)

표면(15강)

고체화학의 응용(16강)

시험 문제

총 일곱 문제

개념을 묻는 문제: 네 문제

공식을 유도하는 문제: 세 문제

소재물리화학의 구성

고전 열역학

열역학 복습

상 전이

통계역학

기초 이론

이상 기체

상호작용 계

고체

고체의 구조

고체의 진동

고체의 전기적 성질

고체의 자기적 성질

표면화학

~~나노화학~~ 응용

연성물질

액체

유리

고분자의 구조

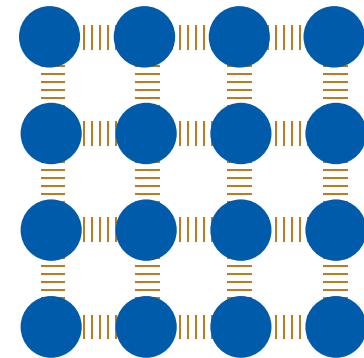
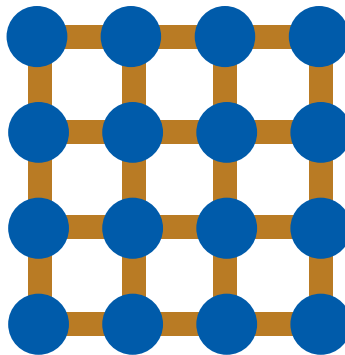
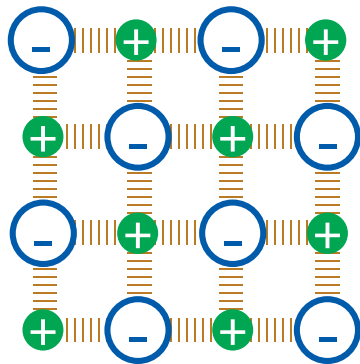
고분자의 상 거동

고분자의 젤화

액정, 콜로이드, ...

Bonding in Solids

- Ionic solids: ionic bonds
- Covalent solids: covalent bonds
- Molecular solids: intermolecular interactions



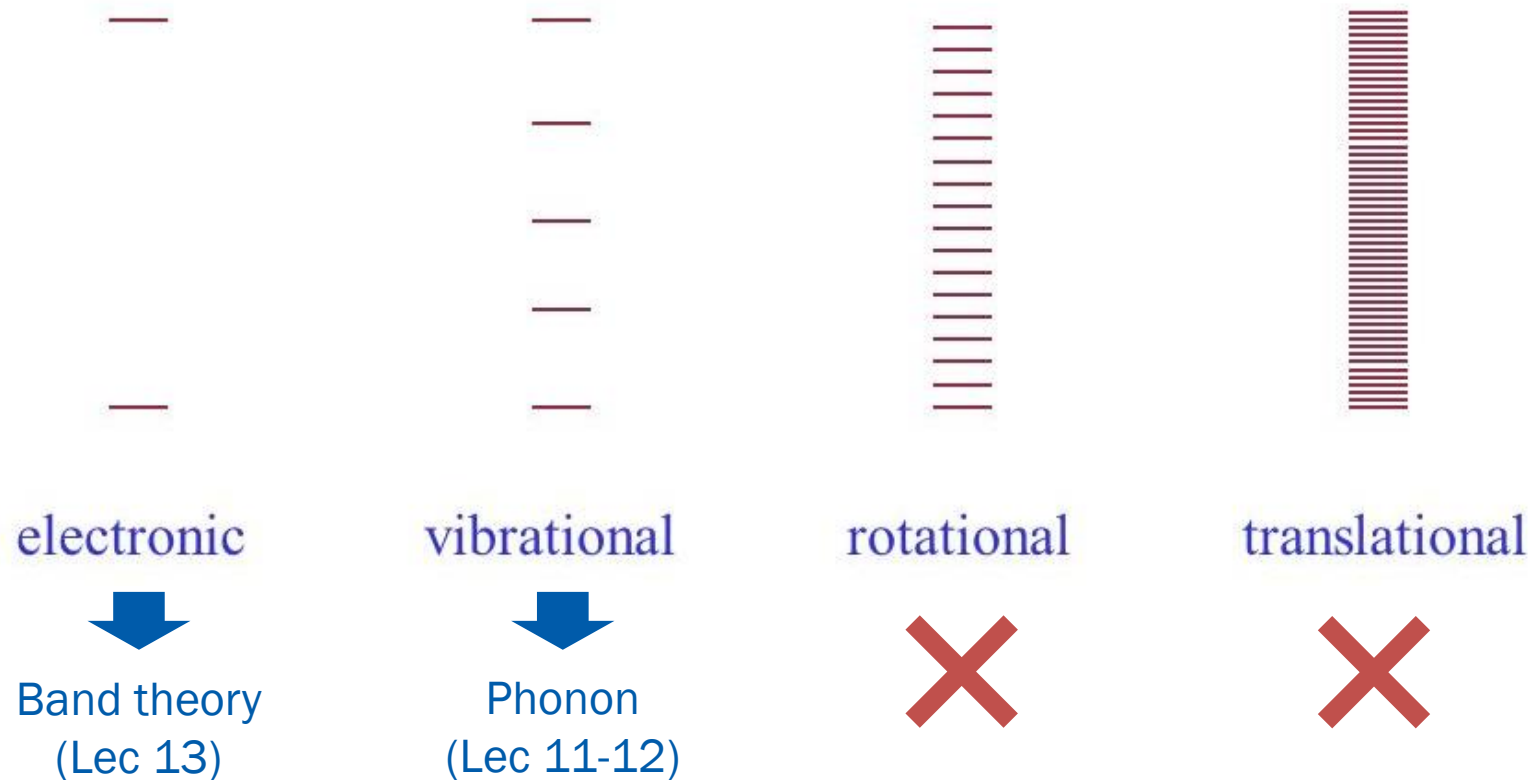
Metals need more discussion

Structures of Solids

- Unit cell
- Primitive unit cell
- Lattice constants
- Lattice systems, Bravais lattices
- Crystal symmetry
- Close packing: cubic (CCP) and hexagonal (HCP)
- Quasicrystals

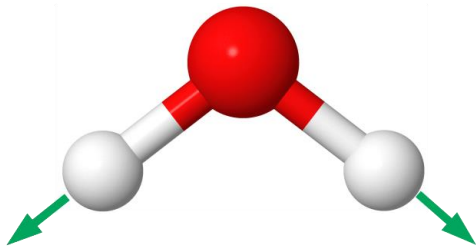
Molecules and Solids

Molecular energies:

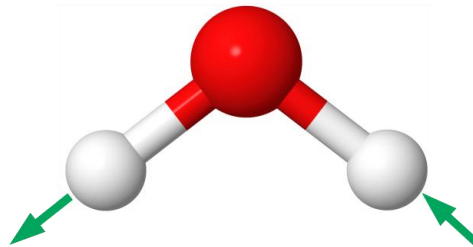


Normal Modes

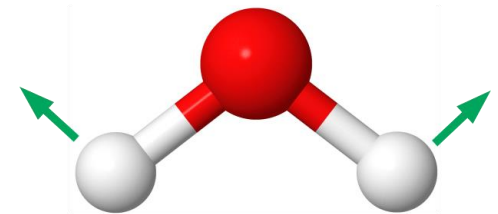
a pattern of motion in which all atoms move with the same frequency



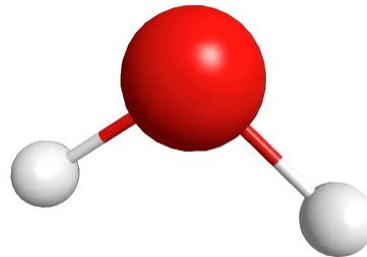
symmetric stretching



asymmetric stretching

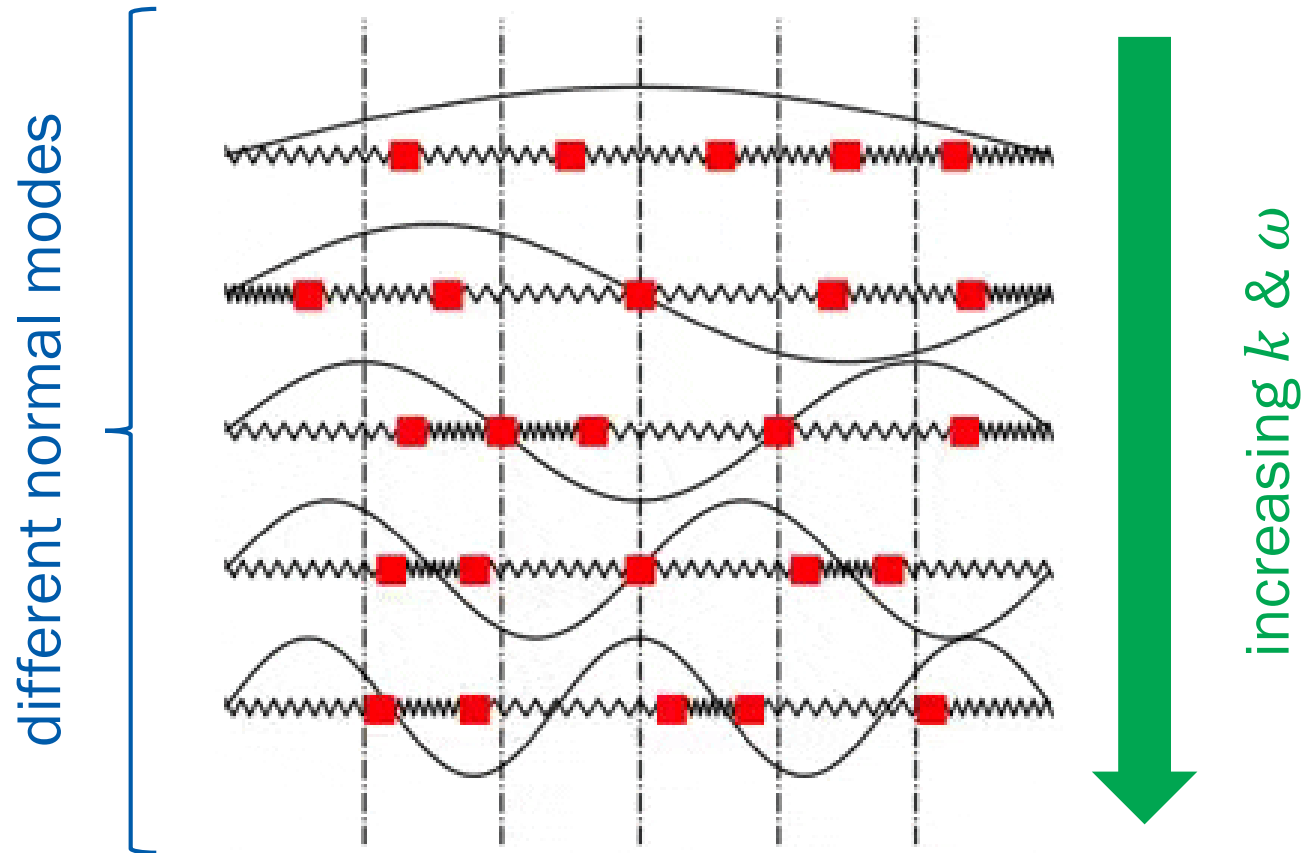


bending



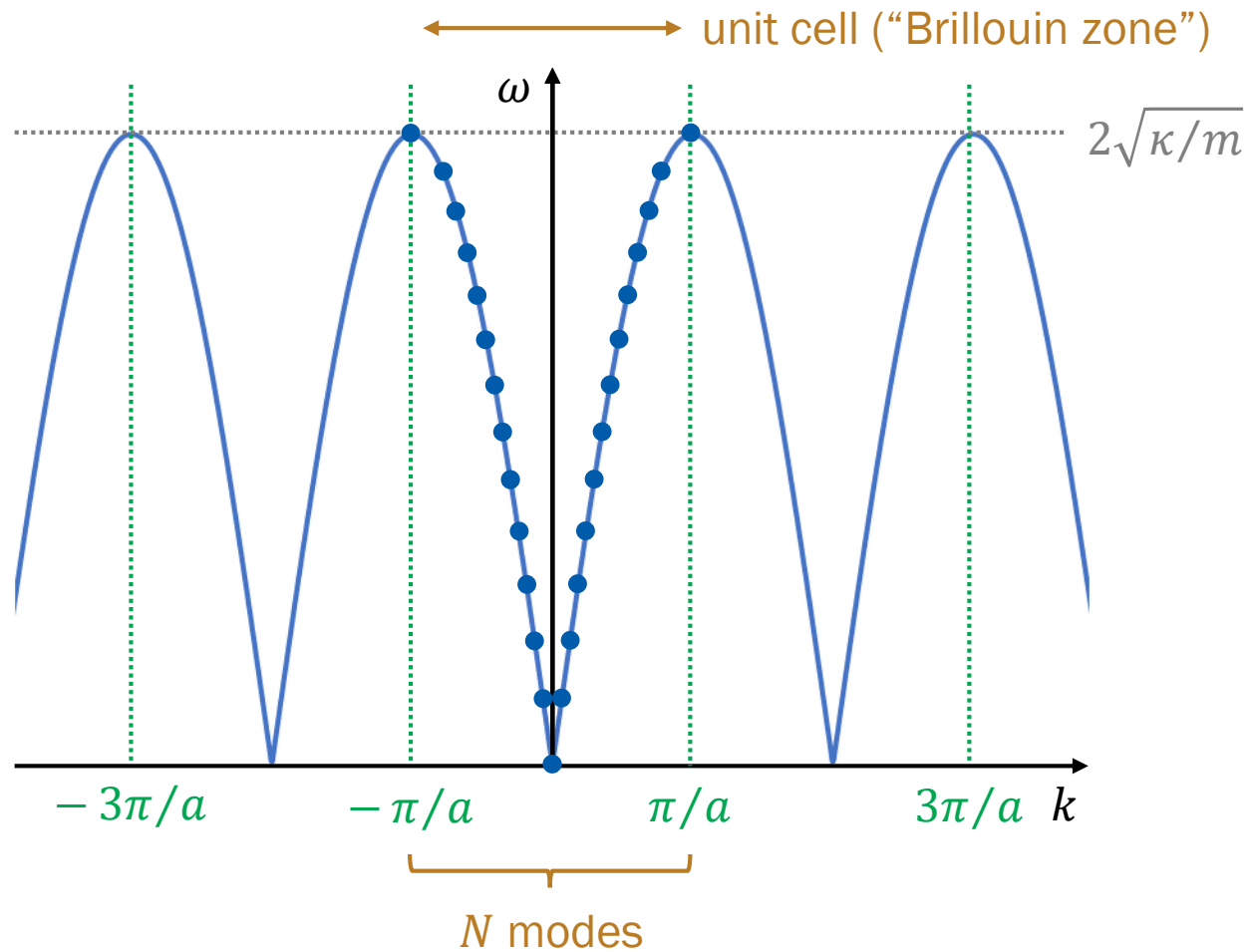
(Images: https://strawberryfields.ai/phononics/_images/water.png
<https://thepaperccny.com/2022/02/28/harmony-of-the-ocean/>)

Lattice Normal Modes ($N = 5$)

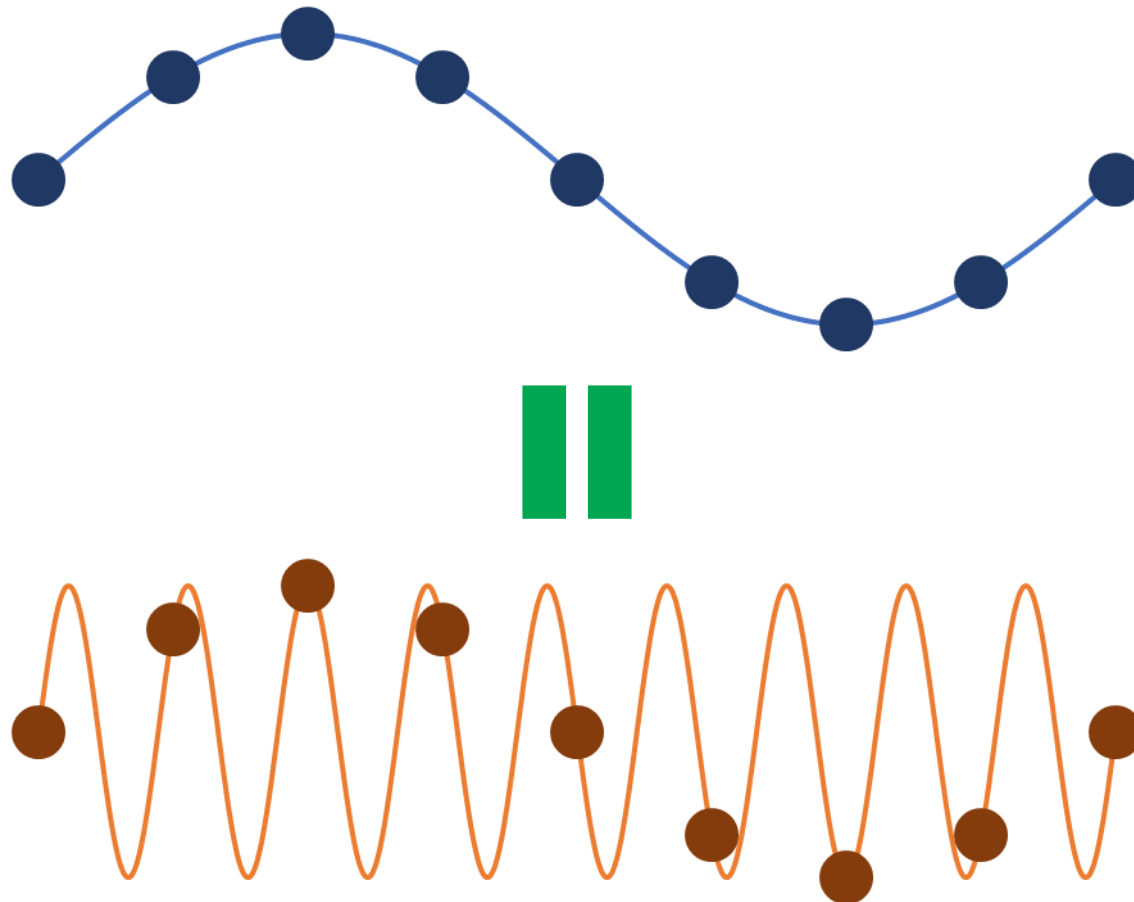


(Image: <https://www.quora.com/What-is-an-intuitive-explanation-of-phonons>)

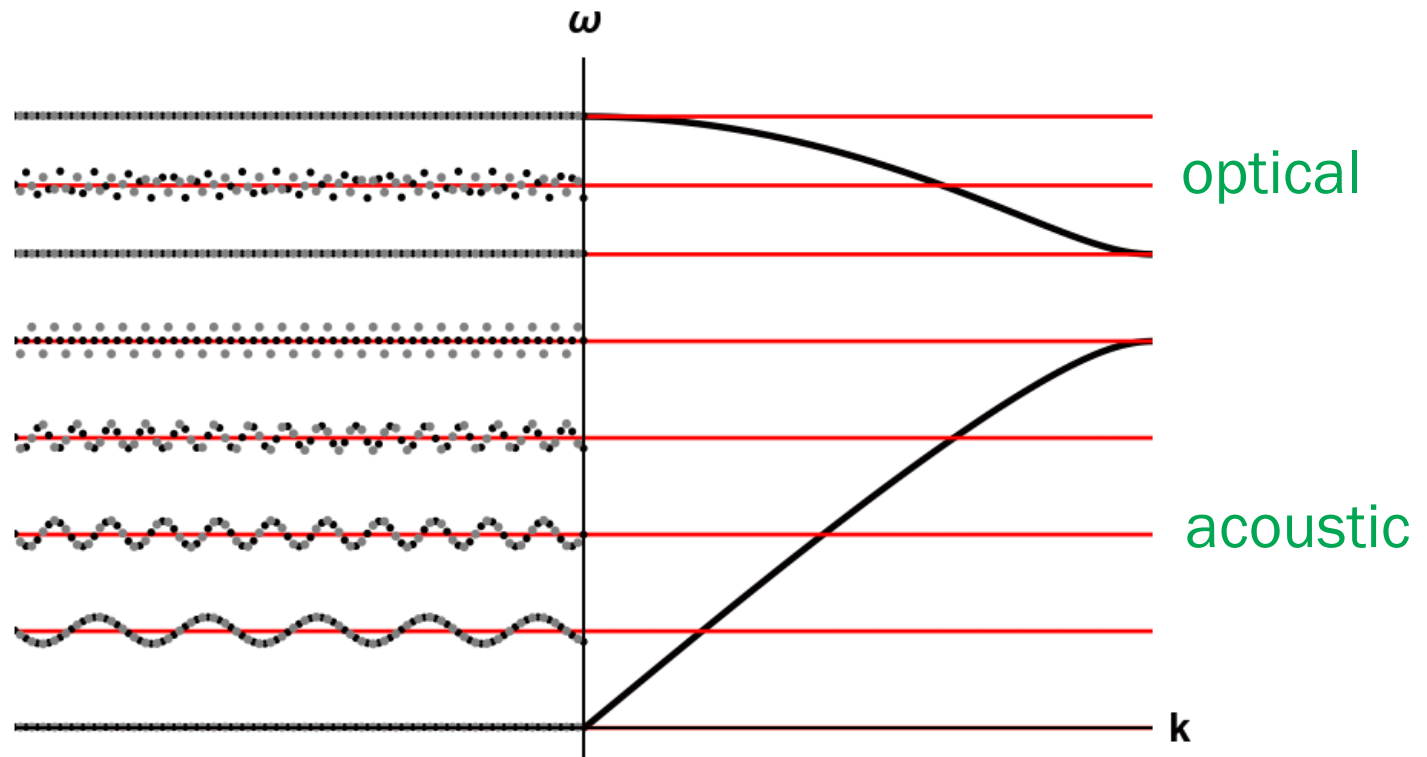
For Large N



Higher-Order Brillouin Zones?



Diatomic Solids



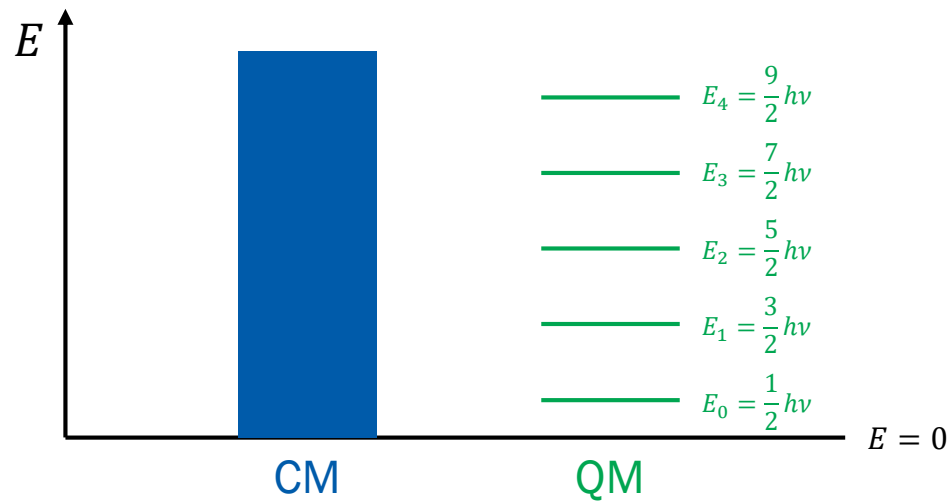
(Image: https://upload.wikimedia.org/wikipedia/commons/d/d7/Diatomic_chain.gif)

Stat Mech of Crystal Vibration

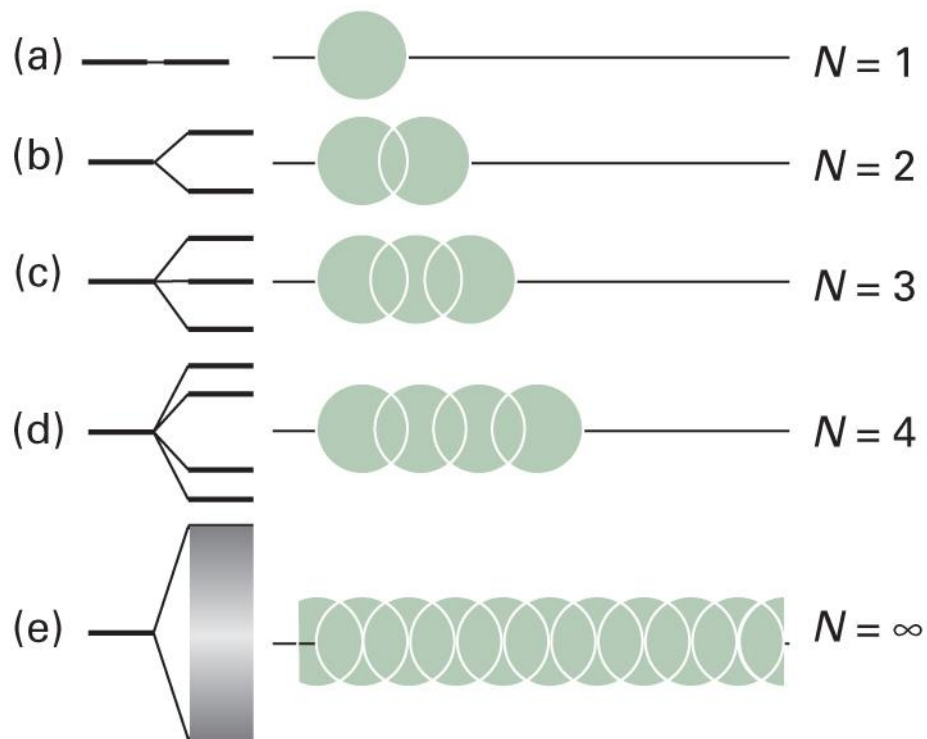
	Identical oscillators	Normal modes
Classical	Dulong-Petit	Dulong-Petit
Quantum	Einstein model	Debye model

“Phonon”

A **phonon** is a discrete **quantum of vibration**
(bosonic)



Electronic Structures of Solids

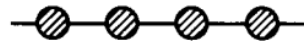


(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 15C.6)

Bands

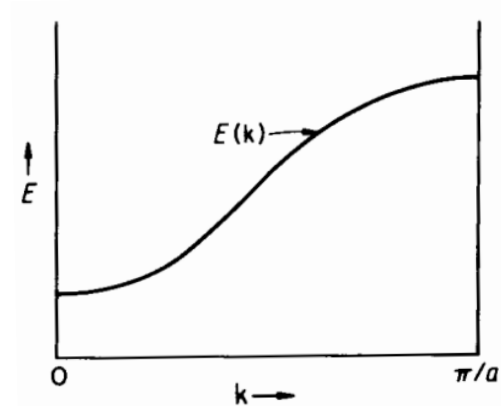
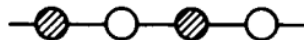
$$\psi_0 = \sum_n e^{i0} \chi_n = \sum_n \chi_n$$

$$= \chi_0 + \chi_1 + \chi_2 + \chi_3 + \dots$$



$$\psi_{\frac{\pi}{a}} = \sum_n e^{i\pi n} \chi_n = \sum_n (-1)^n \chi_n$$

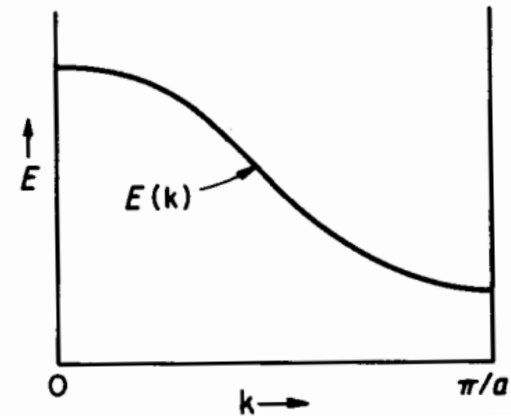
$$= \chi_0 - \chi_1 + \chi_2 - \chi_3 + \dots$$



$$\psi_0 = \chi_0 + \chi_1 + \chi_2 + \chi_3 + \dots$$

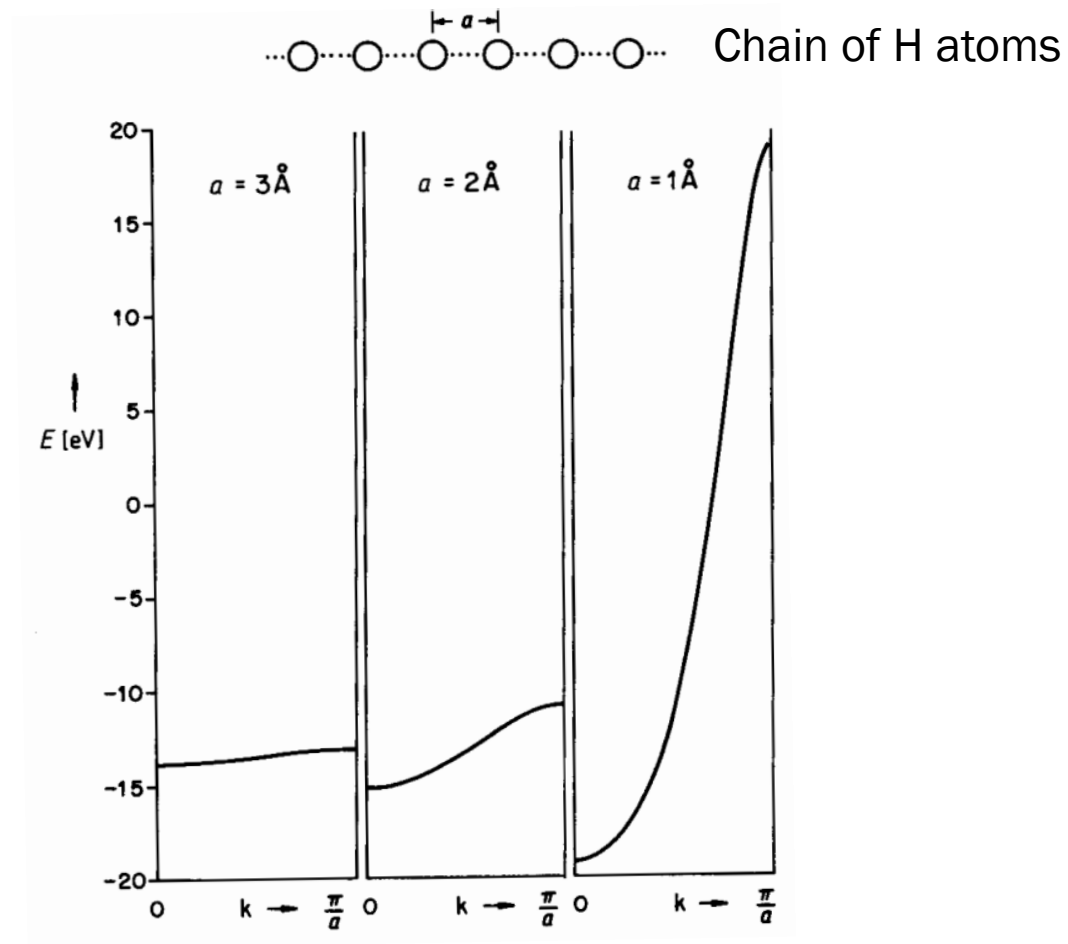


$$\psi_{\frac{\pi}{a}} = \chi_0 - \chi_1 + \chi_2 - \chi_3 + \dots$$



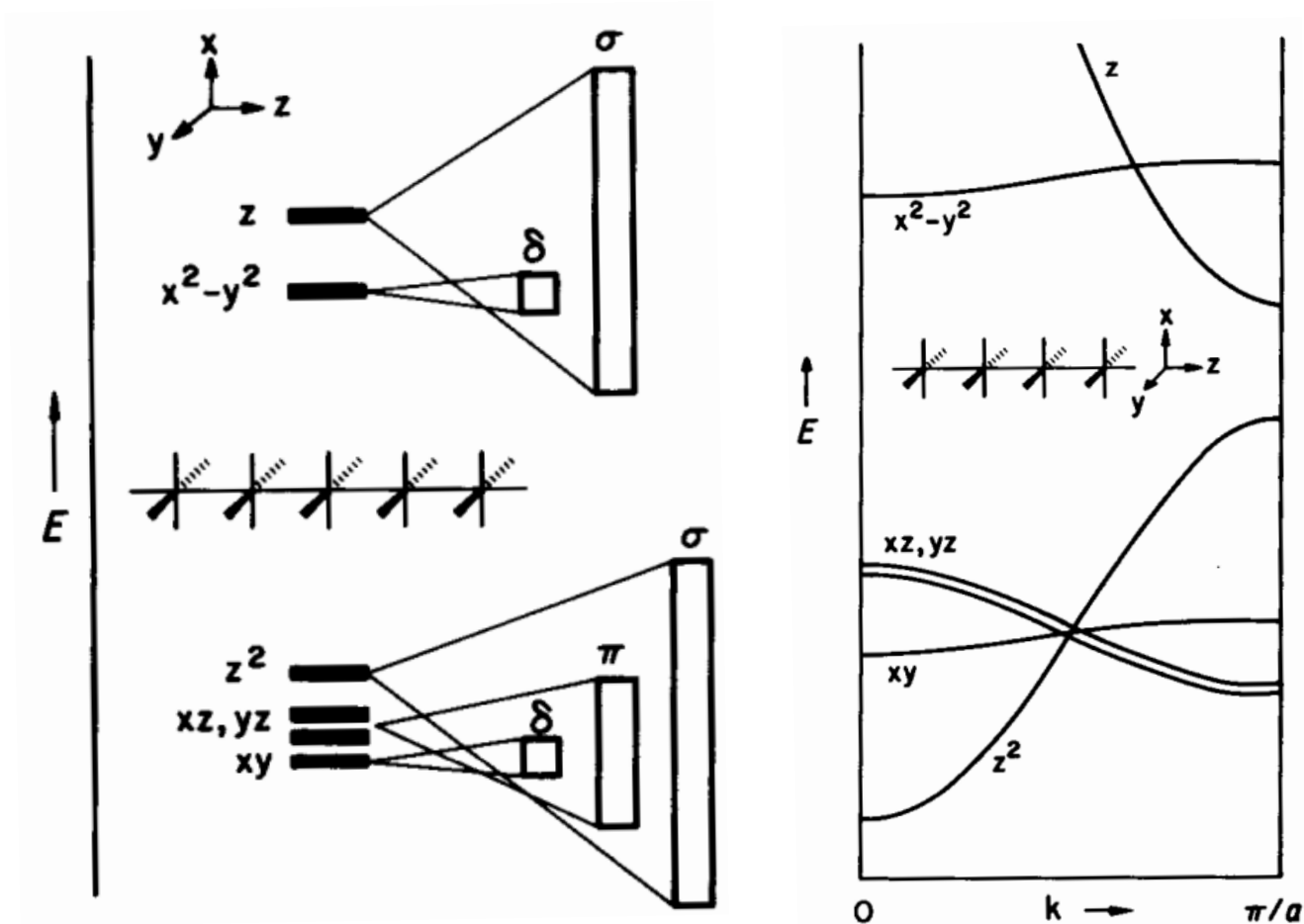
(Image: Hoffmann, *Solids and Surfaces*, p. 7 and p. 9)

“Band Width”



(Image: Hoffmann, *Solids and Surfaces*, Fig. 1)

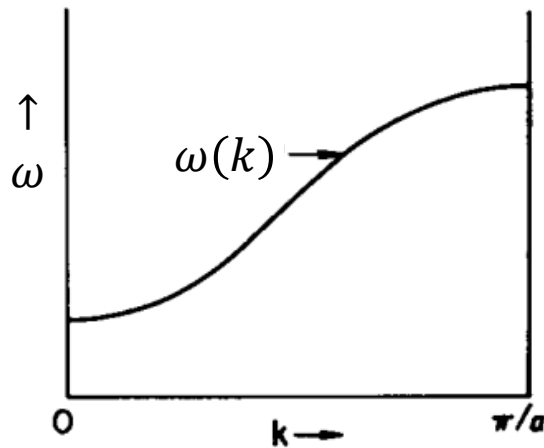
Band Structure



(Image: Hoffmann, *Solids and Surfaces*, p. 13)

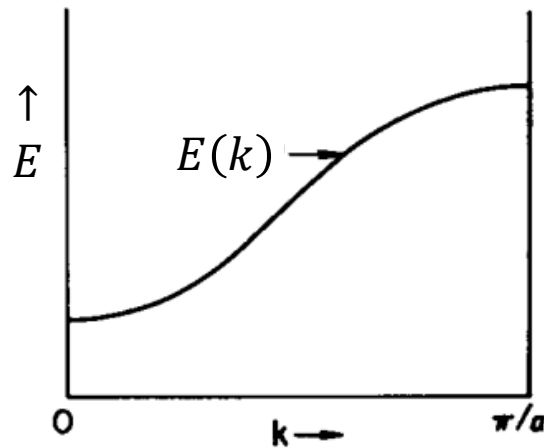
Dispersion Relations

Phonon



modes

Electron

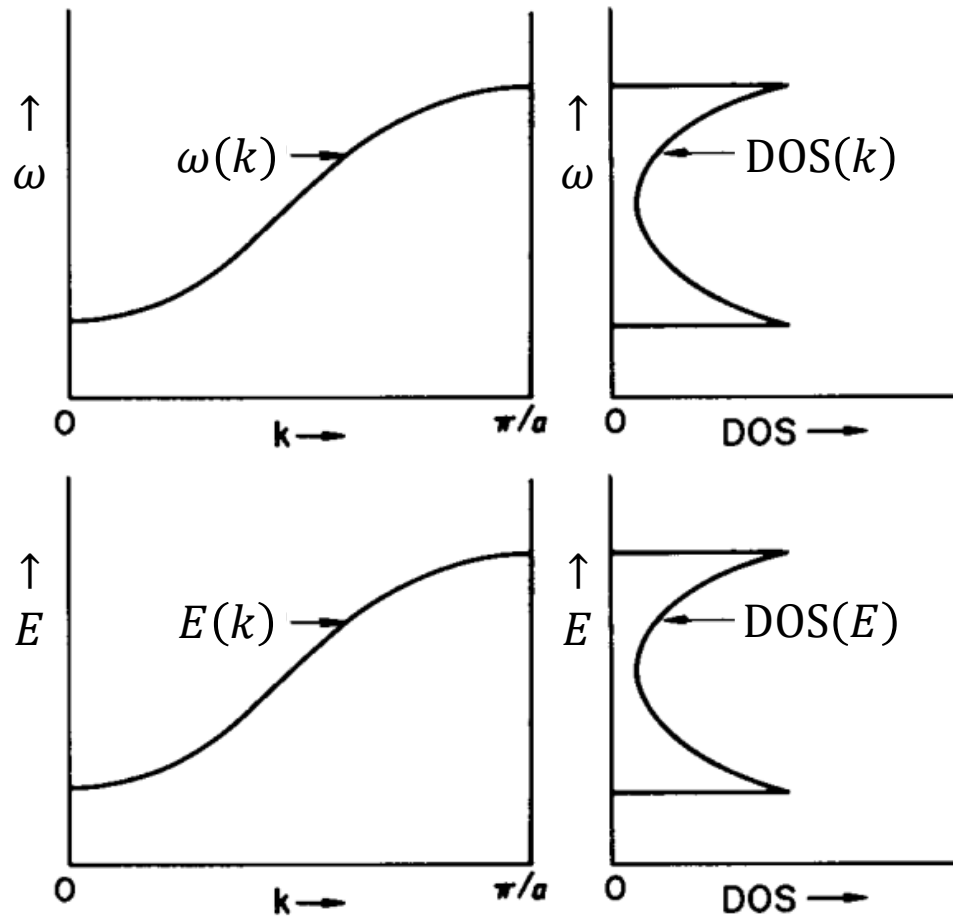


states

(Image: Hoffmann, *Solids and Surfaces*, p. 27)

DOS (Density of States)

Phonon



modes

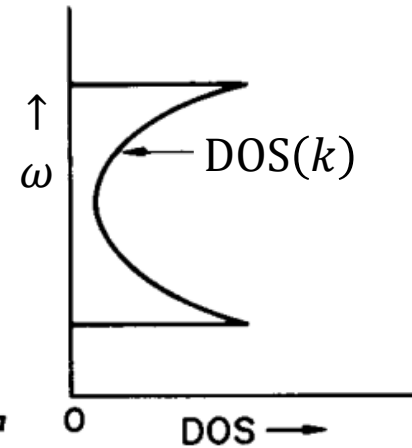
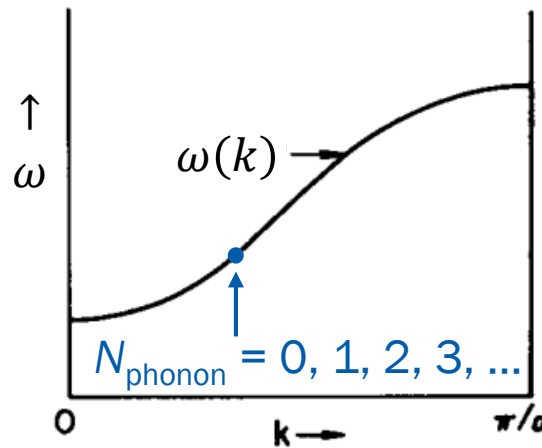
Electron

states

(Image: Hoffmann, *Solids and Surfaces*, p. 27)

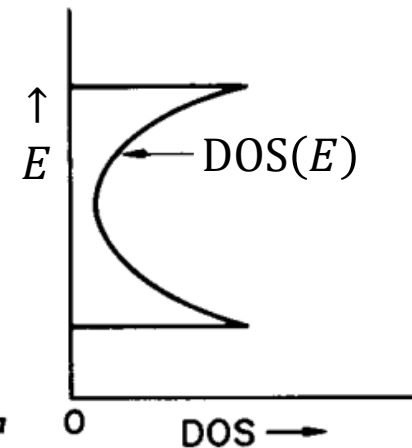
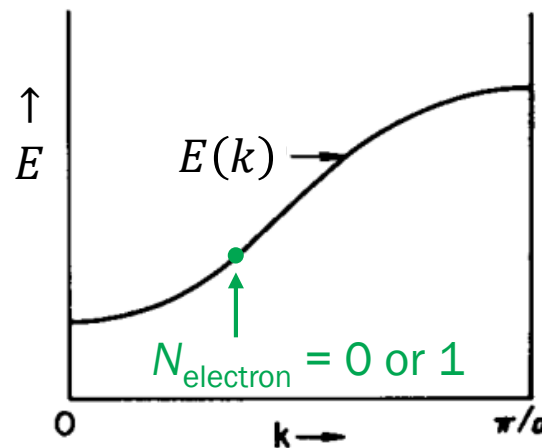
How to Fill States

Phonon



modes

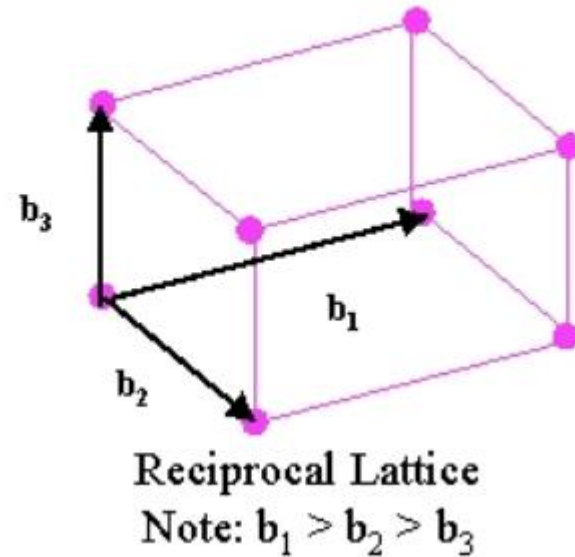
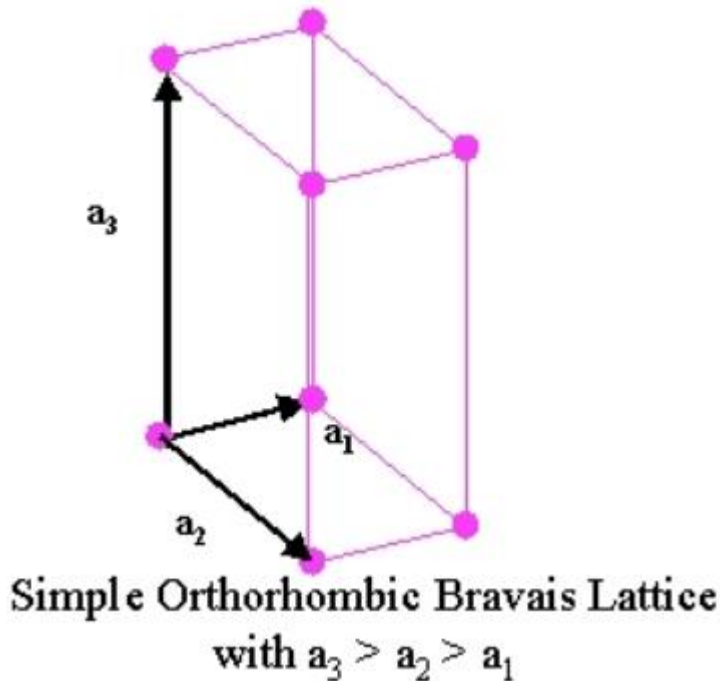
Electron



states

(Image: Hoffmann, *Solids and Surfaces*, p. 27)

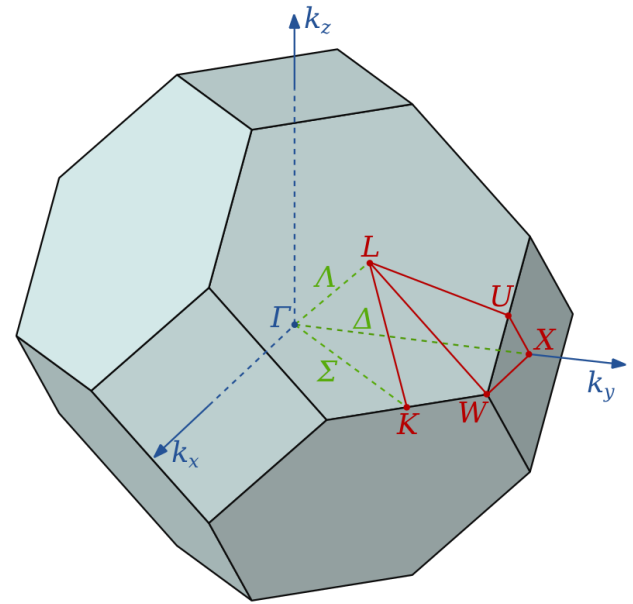
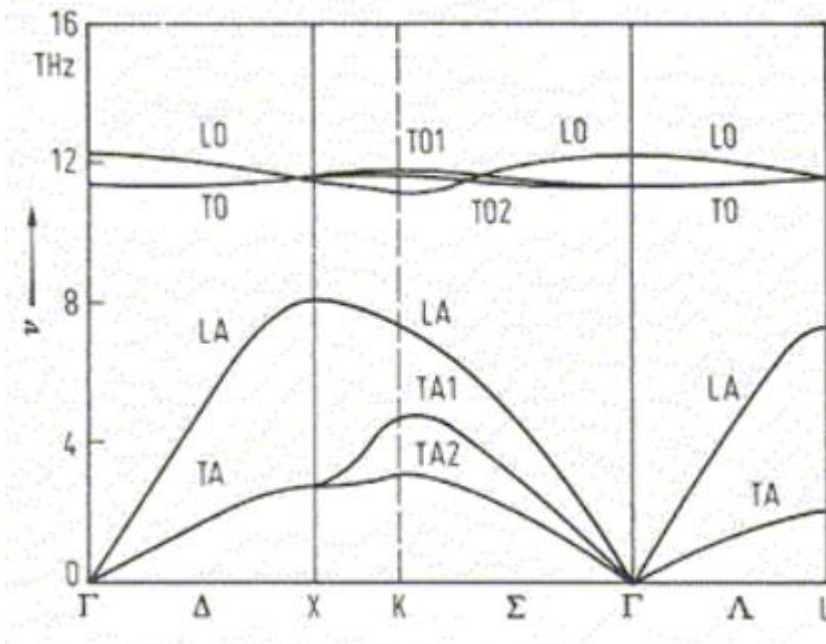
3D Reciprocal Lattice



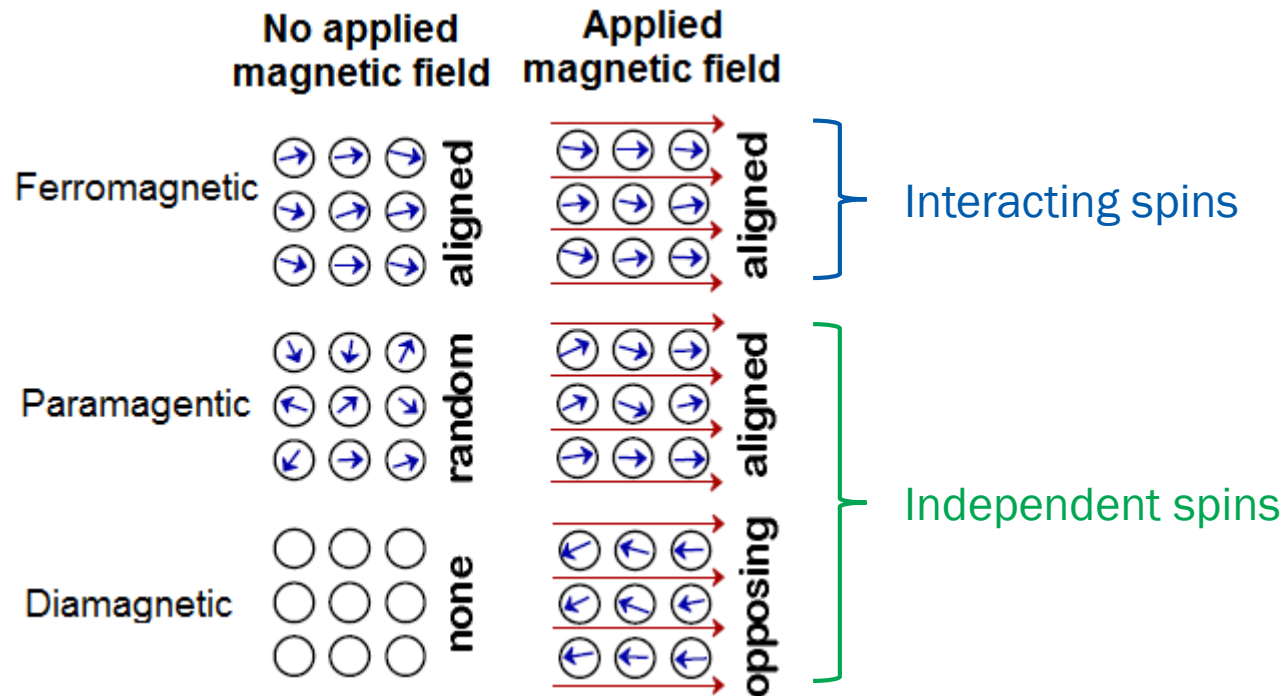
$$\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$$

(Image: <https://www.physics.smu.edu/scalise/P6338sp16/albert/chapter2.pdf>)

Dispersion Relation in 3D



Magnetism



Ising Model

Consideration of neighboring spin-spin interactions

$$E = \mu_B B \sum_i \sigma_i - J \sum_{\langle ij \rangle} \sigma_i \sigma_j$$

- Mean-field approach: exact for $D \geq 4$
- 1D Ising chain

Surfaces (Lec 15)

- Notation: pp. 9-13
- Characterization: pp. 14-19
- Real crystal structures
 - Rearrangement: pp. 21-24
 - Defects: pp. 25-26
- Dispersion relations: p. 27
- Adsorption: pp. 28-41
 - Kinetic models



Applications (Lec 16)

- Defects and doping: pp. 10-14
- Band bending: pp. 15-23
- Nanomaterials: pp. 24-34
 - DOS of free particles